

Adult Health Nursing – I (AHN 1)

Detailed Syllabus Study Notes

Maharashtra University of Health Sciences (MUHS)

BSc Nursing – Academic Cycle 2026

Comprehensive Clinical Study Guide
for BSc Nursing Students

Prepared for academic use only. Verify clinical protocols with current institutional guidelines.

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Unit 1: Introduction to Medical-Surgical Nursing & Clinical Pathophysiology

1.1 Definition and Scope

Medical-surgical nursing is a specialty area that cares for adult patients with medical conditions or those recovering from surgery. It focuses on promoting health, preventing illness, and managing acute and chronic diseases across the lifespan.

Scope includes: assessment, diagnosis, planning, implementation, and evaluation of nursing care; medication administration; patient education; rehabilitation support; and coordination of multidisciplinary care.

1.2 Trends in Medical-Surgical Nursing

Current trends include: evidence-based practice, patient-centered care, use of technology (e.g., electronic health records, telehealth), minimally invasive procedures, early ambulation, and enhanced recovery after surgery (ERAS) protocols.

Nurses now play a larger role in discharge planning, health promotion, chronic disease management, and community-based care.

1.3 Nursing Process in Medical-Surgical Settings

Assessment: collect subjective and objective data, health history, physical examination, and diagnostic findings.

Nursing Diagnosis: identify actual and potential health problems using standardized terminology (e.g., NANDA-I).

Planning: set SMART goals and select evidence-based interventions.

Implementation: perform nursing actions, administer medications, provide education, and coordinate care.

Evaluation: compare patient outcomes with expected goals and revise the care plan as needed.

1.4 Common Pathophysiological Concepts

Inflammation: protective response to injury characterized by redness, heat, swelling, pain, and loss of function. Acute inflammation is short-term; chronic inflammation can lead to tissue damage.

Infection: invasion by microorganisms causing disease. The chain of infection includes infectious agent, reservoir, portal of exit, mode of transmission, portal of entry, and susceptible host.

Immunity and healing: innate and adaptive immunity, wound healing by primary, secondary, and tertiary intention; factors affecting healing (nutrition, circulation, age, infection).

Fluid and electrolyte balance: homeostasis of sodium, potassium, calcium, magnesium, and chloride; acid-base balance (pH, PaCO₂, HCO₃⁻).

Shock: inadequate tissue perfusion leading to cellular hypoxia. Types include hypovolemic, cardiogenic, distributive (septic, anaphylactic, neurogenic), and obstructive shock.

1.5 Nursing Care Plan: Preoperative Anxiety

Assessment	Patient reports worry, increased heart rate, sweating, difficulty sleeping before surgery.
Diagnosis	Anxiety related to impending surgery and unknown outcomes.
Interventions	Provide preoperative teaching; explain procedures in simple language; answer questions; use relaxation techniques; administer anxiolytics as prescribed.
Rationale	Education reduces fear, improves cooperation, and prepares the patient physiologically and psychologically for surgery.

Unit 2: Perioperative Nursing Care

2.1 Preoperative Phase

Preoperative nursing begins when the decision for surgery is made and continues until the patient is transferred to the operating room.

Key activities: obtain informed consent, complete health assessment, review medications and allergies, perform diagnostic tests (CBC, electrolytes, ECG, chest X-ray), ensure NPO status, prepare skin and bowel, and provide psychological support.

Informed consent: voluntary, written permission after explaining the procedure, risks, benefits, and alternatives. The patient must be competent and able to understand.

NPO (nil per os) guidelines: usually 6-8 hours for solids and 2 hours for clear liquids before anesthesia to reduce aspiration risk.

2.2 Intraoperative Phase

Begins when the patient enters the operating room and ends with transfer to the post-anesthesia care unit (PACU).

Roles of the nurse: scrub nurse (sterile field, instruments), circulating nurse (non-sterile, patient advocacy, documentation, supplies), and anesthesia support.

Aseptic technique: hand hygiene, sterile gowns and gloves, sterile field maintenance, and proper surgical site preparation to prevent surgical site infections.

Anesthesia types: general, regional (spinal, epidural), local, and monitored anesthesia care (MAC). The nurse monitors vital signs, airway, and positioning.

2.3 Postoperative Phase

Postoperative care focuses on recovery from anesthesia, pain control, wound healing, prevention of complications, and early mobilization.

Immediate post-op monitoring in PACU: airway, breathing, circulation (ABCs), vital signs, oxygen saturation, level of consciousness, pain, nausea, and surgical site drainage.

Common complications: hemorrhage, infection, deep vein thrombosis (DVT), pulmonary embolism, atelectasis, pneumonia, urinary retention, paralytic ileus, and wound dehiscence.

Pain management: multimodal analgesia (opioids, NSAIDs, regional blocks), non-pharmacological methods (positioning, ice, relaxation), and regular pain assessment.

Early ambulation, incentive spirometry, leg exercises, adequate hydration, and wound care promote recovery.

2.4 Nursing Care Plan: Postoperative Pain

Assessment

Patient reports pain score 7/10 at surgical site; guarding behavior; increased heart rate and blood pressure.

Diagnosis

Acute pain related to surgical incision and tissue trauma.

Interventions	Assess pain using a standardized scale; administer analgesics as prescribed; position comfortably; provide splinting support; teach relaxation techniques; monitor for side effects of opioids.
Rationale	Effective pain control reduces stress, improves respiratory function, promotes early mobilization, and prevents complications such as atelectasis and DVT.

Unit 3: Nursing Management of Patients with Respiratory Disorders

3.1 Pneumonia

Definition: infection of the lung parenchyma caused by bacteria, viruses, or fungi, leading to inflammation and consolidation of alveoli.

Clinical features: fever, productive cough (rust-colored or purulent sputum), chest pain, dyspnea, tachypnea, crackles on auscultation, and fatigue.

Diagnosis: chest X-ray showing infiltrates, sputum culture, CBC, blood culture, and pulse oximetry.

Management: antibiotics (for bacterial pneumonia), oxygen therapy, hydration, antipyretics, bronchodilators, and respiratory physiotherapy.

Nursing care: monitor vital signs and oxygen saturation; administer antibiotics on time; encourage deep breathing and coughing; position patient in semi-Fowler's; maintain fluid intake; provide nutrition.

3.2 Chronic Obstructive Pulmonary Disease (COPD)

Definition: progressive, partially reversible airflow limitation characterized by chronic bronchitis and emphysema. Major risk factor is cigarette smoking.

Clinical features: chronic cough, sputum production, progressive dyspnea, barrel chest, wheezing, and use of accessory muscles.

Diagnosis: spirometry showing $FEV_1/FVC < 0.70$ after bronchodilator; chest X-ray; arterial blood gas (ABG) analysis.

Management: smoking cessation, bronchodilators (salbutamol, ipratropium), inhaled corticosteroids, oxygen therapy (low-flow, long-term if indicated), pulmonary rehabilitation, and vaccination (influenza, pneumococcal).

Nursing care: teach correct inhaler technique; monitor breathing pattern and ABG; administer oxygen at prescribed flow; encourage pursed-lip breathing; small frequent meals; prevent infections.

3.3 Asthma

Definition: chronic inflammatory disorder of the airways causing reversible bronchospasm, mucosal edema, and mucus hypersecretion.

Clinical features: episodic dyspnea, wheezing, chest tightness, cough (especially at night or early morning), and tachypnea during attacks.

Triggers: allergens, respiratory infections, cold air, exercise, stress, smoke, and certain medications (aspirin, beta-blockers).

Management: relievers (short-acting beta-agonists like salbutamol) and controllers (inhaled corticosteroids, long-acting beta-agonists, leukotriene modifiers).

Nursing care: assess respiratory status and peak expiratory flow; administer bronchodilators promptly; position upright; maintain calm environment; teach trigger avoidance and action plan.

3.4 Pleural Effusion

Definition: accumulation of fluid in the pleural space, which can impair lung expansion and cause respiratory distress.

Types: transudate (heart failure, cirrhosis, nephrotic syndrome) and exudate (infection, malignancy, trauma).

Clinical features: dyspnea, chest pain, diminished breath sounds, dullness on percussion, and decreased chest expansion on affected side.

Diagnosis: chest X-ray, ultrasound, CT scan, and thoracentesis for fluid analysis.

Management: treat underlying cause; therapeutic thoracentesis; chest tube drainage for large or recurrent effusions; analgesics.

Nursing care: monitor respiratory effort and oxygen saturation; assist with thoracentesis; maintain chest tube drainage system; position patient on affected side to improve ventilation of healthy lung.

3.5 Tuberculosis (TB)

Definition: infectious disease caused by *Mycobacterium tuberculosis*, primarily affecting the lungs but capable of spreading to other organs.

Clinical features: persistent cough > 2-3 weeks, hemoptysis, low-grade fever, night sweats, weight loss, fatigue, and anorexia.

Diagnosis: sputum acid-fast bacilli (AFB) smear, culture, GeneXpert MTB/RIF, chest X-ray, and Mantoux test (Tuberculin skin test).

Management: Directly Observed Therapy Short-course (DOTS) with first-line drugs: Isoniazid, Rifampicin, Pyrazinamide, Ethambutol (HRZE) for 2 months intensive phase, followed by HR for 4 months continuation phase.

Nursing care: educate about medication adherence and side effects; monitor liver function; ensure respiratory isolation until sputum is negative; provide nutrition support; contact tracing and screening.

3.6 Nursing Care Plan: Impaired Gas Exchange

Assessment	Dyspnea, cyanosis, restlessness, abnormal ABG (low PaO ₂ , high PaCO ₂), decreased SpO ₂ .
Diagnosis	Impaired gas exchange related to ventilation-perfusion mismatch secondary to lung disease.
Interventions	Administer oxygen as prescribed; monitor SpO ₂ and ABG; position in semi-Fowler's or high Fowler's; encourage deep breathing and coughing; suction as needed; administer bronchodilators and antibiotics; limit activity.
Rationale	Optimizing oxygenation and ventilation improves tissue perfusion, reduces hypoxemia, and prevents respiratory failure.

Unit 4: Nursing Management of Patients with Cardiovascular & Hematological Disorders

4.1 Hypertension

Definition: persistent elevation of blood pressure $\geq 140/90$ mmHg. Primary (essential) hypertension has no identifiable cause; secondary hypertension results from kidney disease, endocrine disorders, or medications.

Clinical features: often asymptomatic in early stages; severe hypertension may cause headache, dizziness, chest pain, dyspnea, and visual disturbances.

Risk factors: family history, obesity, sedentary lifestyle, high salt intake, smoking, alcohol, stress, diabetes, and chronic kidney disease.

Management: lifestyle modifications (DASH diet, exercise, weight loss, smoking cessation, limit alcohol); antihypertensive drugs (ACE inhibitors, ARBs, calcium channel blockers, diuretics, beta-blockers).

Nursing care: monitor BP regularly; educate about medication adherence and side effects; encourage low-sodium diet; stress management; monitor for complications (stroke, MI, heart failure, kidney disease).

4.2 Coronary Artery Disease (CAD) and Myocardial Infarction (MI)

CAD: narrowing of coronary arteries due to atherosclerosis, reducing blood flow to the myocardium. MI occurs when blood supply is severely reduced or blocked, causing myocardial necrosis.

Clinical features of MI: crushing chest pain radiating to left arm/jaw/back, diaphoresis, nausea, dyspnea, anxiety, and palpitations. Some patients (especially women, diabetics, elderly) may have atypical symptoms.

Diagnosis: ECG (ST elevation, T-wave inversion, pathological Q waves), cardiac troponins, CK-MB, echocardiography, and coronary angiography.

Management: MONA approach (Morphine, Oxygen, Nitrates, Aspirin) for MI; thrombolysis or PCI (percutaneous coronary intervention) for STEMI; beta-blockers, statins, antiplatelets, ACE inhibitors.

Nursing care: continuous cardiac monitoring; pain assessment and management; administer oxygen, nitrates, antiplatelets; monitor for arrhythmias, heart failure, and cardiogenic shock; emotional support and cardiac rehabilitation education.

4.3 Heart Failure

Definition: clinical syndrome where the heart cannot pump enough blood to meet the body's metabolic needs. Classified as left-sided or right-sided, and systolic or diastolic.

Clinical features: left-sided failure causes pulmonary congestion (dyspnea, orthopnea, paroxysmal nocturnal dyspnea, cough, pink frothy sputum); right-sided failure causes systemic congestion (peripheral edema, hepatomegaly, ascites, jugular venous distension).

Diagnosis: echocardiography, BNP/NT-proBNP, chest X-ray, ECG, and ABG.

Management: diuretics, ACE inhibitors/ARBs, beta-blockers, aldosterone antagonists, vasodilators, inotropes in acute cases, and device therapy (CRT, ICD) when indicated.

Nursing care: monitor intake and output, daily weights, edema, and vital signs; administer diuretics and monitor electrolytes; position in semi-Fowler's or high Fowler's; sodium and fluid restriction; activity pacing; medication education.

4.4 Anemia

Definition: reduction in hemoglobin, red blood cell count, or hematocrit below normal levels, leading to decreased oxygen-carrying capacity.

Common types: iron deficiency anemia, megaloblastic anemia (B12/folate deficiency), hemolytic anemia, and anemia of chronic disease.

Clinical features: fatigue, pallor, weakness, dyspnea on exertion, tachycardia, dizziness, and glossitis (in B12/folate deficiency).

Diagnosis: CBC, peripheral smear, serum iron, ferritin, TIBC, vitamin B12, folate, and reticulocyte count.

Management: treat underlying cause; iron supplements for iron deficiency; B12/folate supplementation; blood transfusion in severe symptomatic anemia; erythropoietin in renal failure.

Nursing care: assess fatigue and activity tolerance; administer iron with vitamin C; monitor response to therapy; provide high-iron foods; teach about side effects (constipation, black stools with iron); safety measures to prevent falls.

4.5 Nursing Care Plan: Decreased Cardiac Output

Assessment	Tachycardia, hypotension, weak pulse, dyspnea, reduced urine output, fatigue, cool clammy skin.
Diagnosis	Decreased cardiac output related to impaired myocardial contractility or increased afterload.
Interventions	Monitor vital signs, ECG, and urine output; administer oxygen, diuretics, inotropes, and vasodilators as prescribed; position patient upright; limit activity; maintain accurate intake and output; daily weight.
Rationale	Improving cardiac output enhances tissue perfusion, reduces pulmonary congestion, and prevents organ dysfunction.

Unit 5: Nursing Management of Patients with Gastrointestinal & Digestive Disorders

5.1 Peptic Ulcer Disease

Definition: erosion of the mucosal lining of the stomach or duodenum caused by *Helicobacter pylori* infection, NSAID use, stress, smoking, or excessive alcohol.

Clinical features: burning epigastric pain, pain relieved by food (duodenal ulcer) or worsened by food (gastric ulcer), nausea, bloating, hematemesis, or melena if bleeding occurs.

Diagnosis: upper GI endoscopy, H. pylori testing (urea breath test, stool antigen, biopsy), and barium swallow.

Management: H. pylori eradication therapy (triple/quadruple therapy), proton pump inhibitors (PPIs), H2 blockers, antacids, and avoidance of NSAIDs, alcohol, and smoking.

Nursing care: administer medications on schedule; monitor for signs of bleeding; teach dietary modifications (small frequent meals, avoid spicy/acidic foods); stress management; educate about medication adherence.

5.2 Hepatitis

Definition: inflammation of the liver, commonly caused by viral infections (Hepatitis A, B, C, D, E), alcohol, drugs, or toxins.

Clinical features: jaundice, dark urine, pale stools, fatigue, anorexia, nausea, right upper quadrant pain, hepatomegaly, and pruritus.

Diagnosis: liver function tests (ALT, AST, bilirubin, alkaline phosphatase), viral serology (HBsAg, anti-HCV), ultrasound, and sometimes liver biopsy.

Management: rest, hydration, nutrition support; antivirals for hepatitis B and C; avoid alcohol and hepatotoxic drugs; vaccination for hepatitis A and B; monitor for complications (cirrhosis, liver cancer).

Nursing care: monitor liver function and signs of bleeding; provide rest and small frequent meals; educate about transmission prevention, vaccination, and medication safety; standard precautions for blood/body fluids.

5.3 Cirrhosis

Definition: chronic, irreversible scarring of the liver due to long-term damage from hepatitis, alcohol, fatty liver disease, or autoimmune conditions.

Clinical features: jaundice, ascites, peripheral edema, hepatic encephalopathy (confusion, asterix), gynecomastia, spider angiomas, palmar erythema, and bleeding tendencies.

Diagnosis: liver function tests, ultrasound, FibroScan, CT/MRI, endoscopy (for varices), and liver biopsy.

Management: treat underlying cause; diuretics for ascites; lactulose/rifaximin for encephalopathy; beta-blockers for variceal bleeding prophylaxis; nutritional support; liver transplantation in end-stage disease.

Nursing care: monitor fluid balance, abdominal girth, weight, and mental status; restrict sodium and fluid; administer diuretics and lactulose; prevent falls; monitor for bleeding; educate about alcohol abstinence.

5.4 Appendicitis

Definition: inflammation of the appendix, usually due to obstruction by fecalith, lymphoid hyperplasia, or foreign body. It is a common surgical emergency.

Clinical features: colicky periumbilical pain migrating to right lower quadrant (McBurney's point), nausea, vomiting, low-grade fever, rebound tenderness, and leukocytosis.

Diagnosis: clinical examination, ultrasound, CT scan, and CBC showing elevated WBC.

Management: appendectomy (open or laparoscopic); antibiotics for uncomplicated cases in selected patients; IV fluids and analgesics.

Nursing care: preoperative NPO, IV fluids, antibiotics, and pain management; postoperative monitoring for wound infection, peritonitis, ileus; early ambulation; discharge education.

5.5 Nursing Care Plan: Imbalanced Nutrition: Less Than Body Requirements

Assessment	Weight loss, decreased appetite, nausea, weakness, low albumin, poor wound healing.
Diagnosis	Imbalanced nutrition: less than body requirements related to anorexia and impaired digestion.
Interventions	Assess dietary intake and weight; provide small frequent meals high in calories and protein; administer antiemetics; monitor lab values; refer to dietitian; encourage oral hygiene.
Rationale	Adequate nutrition supports healing, immune function, and recovery from GI disorders.

Unit 6: Nursing Management of Patients with Musculoskeletal Disorders

6.1 Fractures

Definition: break in the continuity of a bone caused by trauma, falls, sports injuries, or pathological conditions (e.g., osteoporosis, bone tumors).

Types: closed (simple) and open (compound); complete and incomplete; comminuted, transverse, oblique, spiral, greenstick, and impacted.

Clinical features: pain, swelling, deformity, inability to bear weight, crepitus, bruising, and loss of function.

Emergency management: immobilization, control bleeding, pain relief, and prevention of further injury. Open fractures require tetanus prophylaxis and antibiotics.

Treatment: reduction (closed or open), immobilization (cast, splint, traction), internal fixation (plates, screws, nails), and rehabilitation.

Nursing care: neurovascular assessment (5 Ps: pain, pallor, pulse, paresthesia, paralysis), elevate limb, monitor for compartment syndrome, DVT prophylaxis, cast care, and physiotherapy education.

6.2 Osteoarthritis

Definition: degenerative joint disease characterized by cartilage breakdown, subchondral bone changes, and joint pain. It is the most common form of arthritis.

Clinical features: joint pain worsened by activity and relieved by rest, stiffness (usually <30 minutes), reduced range of motion, crepitus, and bony enlargements (Heberden's/Bouchard's nodes in hands).

Diagnosis: clinical examination and X-ray showing joint space narrowing, osteophytes, subchondral sclerosis, and cysts.

Management: weight reduction, exercise (strengthening and range-of-motion), physiotherapy, analgesics (acetaminophen, NSAIDs), intra-articular corticosteroids, and joint replacement in severe cases.

Nursing care: assist with ADLs; recommend assistive devices; teach joint protection techniques; monitor medication side effects (especially NSAIDs); encourage weight management and regular activity.

6.3 Rheumatoid Arthritis

Definition: chronic autoimmune inflammatory disorder affecting synovial joints, causing symmetric polyarthritis, joint deformity, and systemic symptoms.

Clinical features: morning stiffness > 1 hour, symmetric joint swelling, pain, warmth, ulnar deviation, swan-neck deformity, rheumatoid nodules, fatigue, and low-grade fever.

Diagnosis: rheumatoid factor (RF), anti-CCP antibodies, elevated ESR/CRP, and X-ray/ultrasound showing erosions and joint space narrowing.

Management: disease-modifying antirheumatic drugs (DMARDs) such as methotrexate, hydroxychloroquine, sulfasalazine; biologics (TNF inhibitors); NSAIDs; corticosteroids; physiotherapy;

joint protection.

Nursing care: encourage medication adherence and monitor for side effects (liver toxicity, bone marrow suppression); assist with morning stiffness routines; provide heat/cold therapy; teach joint protection; monitor for infection due to immunosuppression.

6.4 Amputation

Definition: surgical removal of a limb or part of a limb, performed due to trauma, peripheral vascular disease, diabetes, infection, malignancy, or severe congenital anomalies.

Preoperative care: psychological preparation, assess vascular status, nutrition, skin care, and informed consent.

Postoperative care: monitor stump wound and drainage; prevent contractures; position limb properly; pain management (including phantom limb pain); stump wrapping and shaping; early prosthetic fitting and rehabilitation.

Complications: hemorrhage, infection, phantom limb pain, contractures, delayed wound healing, and psychological distress.

Nursing care: monitor vital signs and stump dressing; encourage limb positioning to prevent contractures; provide pain relief; support emotional adjustment; teach stump care and prosthesis use; refer to rehabilitation services.

6.5 Nursing Care Plan: Impaired Physical Mobility

Assessment	Limited range of motion, pain on movement, inability to perform activities independently, use of assistive devices.
Diagnosis	Impaired physical mobility related to musculoskeletal injury or degenerative joint disease.
Interventions	Assist with ROM exercises; provide assistive devices; administer analgesics before activity; encourage gradual ambulation; prevent falls; refer to physiotherapy.
Rationale	Maintaining mobility prevents complications like contractures, DVT, pressure ulcers, and muscle atrophy.

Unit 7: Nursing Management of Patients with Genitourinary & Renal Disorders

7.1 Urinary Tract Infection (UTI)

Definition: infection of any part of the urinary system, most commonly the bladder (cystitis) and urethra (urethritis). Pyelonephritis involves the kidneys.

Clinical features: dysuria, frequency, urgency, hematuria, suprapubic pain, cloudy urine, and fever/flank pain in pyelonephritis.

Risk factors: female anatomy, sexual activity, urinary catheterization, diabetes, pregnancy, and urinary retention.

Diagnosis: urinalysis (pyuria, bacteriuria), urine culture and sensitivity, and ultrasound if recurrent.

Management: antibiotics based on culture (e.g., nitrofurantoin, ciprofloxacin, cephalexin), increased fluid intake, urinary analgesics (phenazopyridine), and treatment of underlying causes.

Nursing care: encourage fluid intake (unless contraindicated); monitor urine output and characteristics; administer antibiotics; teach perineal hygiene; complete full antibiotic course; catheter care if indicated.

7.2 Renal Calculi (Kidney Stones)

Definition: hard mineral and salt deposits formed in the kidneys, which can cause obstruction, pain, and infection as they pass through the urinary tract.

Clinical features: severe flank pain (renal colic) radiating to groin, hematuria, nausea, vomiting, dysuria, and urinary frequency.

Types: calcium oxalate (most common), calcium phosphate, uric acid, struvite, and cystine stones.

Diagnosis: ultrasound, non-contrast CT scan (CT KUB), KUB X-ray, urinalysis, and serum creatinine/uric acid.

Management: hydration, analgesics (NSAIDs), medical expulsive therapy (alpha-blockers like tamsulosin), lithotripsy (ESWL), ureteroscopy, and percutaneous nephrolithotomy (PCNL) for large stones.

Nursing care: strain urine to collect stone; monitor pain and urine output; encourage fluids; monitor for infection and obstruction; post-procedure care after lithotripsy or surgery; dietary teaching based on stone type.

7.3 Acute Kidney Injury (AKI)

Definition: sudden decrease in kidney function over hours to days, leading to accumulation of nitrogenous waste, electrolyte imbalance, and fluid overload.

Causes: prerenal (hypovolemia, heart failure, shock), intrinsic (acute tubular necrosis, glomerulonephritis, nephrotoxic drugs), and postrenal (obstruction).

Clinical features: oliguria or anuria, edema, hypertension, nausea, confusion, electrolyte abnormalities (hyperkalemia), and metabolic acidosis.

Diagnosis: elevated serum creatinine and BUN, decreased urine output, electrolyte abnormalities, and urinalysis findings.

Management: treat underlying cause; fluid management; dialysis if indicated; monitor electrolytes; avoid nephrotoxic drugs; nutritional support.

Nursing care: strict intake and output monitoring; daily weights; electrolyte monitoring; restrict potassium and sodium; dialysis care; medication dosing adjustments; infection prevention.

7.4 Chronic Kidney Disease (CKD) and Dialysis

Definition: progressive, irreversible loss of kidney function over months to years, classified into 5 stages based on GFR.

Clinical features: fatigue, anemia, edema, hypertension, nocturia, pruritus, nausea, neuropathy, and bone disease (renal osteodystrophy).

Diagnosis: eGFR, serum creatinine, BUN, urinalysis, electrolytes, and imaging.

Management: control blood pressure and diabetes; dietary restrictions (low sodium, potassium, phosphorus, protein); erythropoietin for anemia; phosphate binders; vitamin D; and renal replacement therapy when indicated.

Dialysis: hemodialysis (blood filtered through machine) and peritoneal dialysis (dialysate infused into peritoneal cavity).

Nursing care: monitor pre/post dialysis weight, vital signs, electrolytes; vascular access care (AV fistula/graft, catheter); infection prevention; diet and fluid restriction education; psychosocial support.

7.5 Nursing Care Plan: Excess Fluid Volume

Assessment	Edema, weight gain, hypertension, dyspnea, crackles, oliguria, jugular venous distension.
Diagnosis	Excess fluid volume related to compromised renal regulatory mechanisms.
Interventions	Monitor intake and output, daily weights, and edema; restrict sodium and fluid as ordered; administer diuretics; elevate legs; monitor lung sounds and vital signs; prepare for dialysis if indicated.
Rationale	Preventing fluid overload reduces pulmonary edema, hypertension, and cardiac strain.

Unit 8: Nursing Management of Patients with Endocrine Disorders

8.1 Diabetes Mellitus

Definition: chronic metabolic disorder characterized by elevated blood glucose due to insulin deficiency, insulin resistance, or both. Types include Type 1, Type 2, gestational diabetes, and other specific types.

Clinical features: polyuria, polydipsia, polyphagia, weight loss, fatigue, blurred vision, and recurrent infections. Long-term complications affect eyes, kidneys, nerves, and cardiovascular system.

Diagnosis: fasting blood glucose ≥ 126 mg/dL, HbA1c $\geq 6.5\%$, random glucose ≥ 200 mg/dL with symptoms, or oral glucose tolerance test ≥ 200 mg/dL.

Management: lifestyle modifications (diet, exercise, weight control); oral hypoglycemic agents (metformin, sulfonylureas, DPP-4 inhibitors, SGLT2 inhibitors); insulin therapy for Type 1 and advanced Type 2; blood glucose monitoring; prevention of complications.

Acute complications: hypoglycemia (treat with 15 g fast-acting carbohydrate), diabetic ketoacidosis (DKA), and hyperosmolar hyperglycemic state (HHS).

Nursing care: monitor blood glucose; administer insulin and oral agents; teach foot care, diet, and self-monitoring; recognize signs of hypo/hyperglycemia; prevent complications; provide emotional support.

8.2 Thyroid Disorders

Hypothyroidism: deficiency of thyroid hormones causing slowed metabolism. Causes include Hashimoto's thyroiditis, iodine deficiency, surgery, or radiation. Features: fatigue, weight gain, cold intolerance, dry skin, constipation, bradycardia, and depression.

Hyperthyroidism: excess thyroid hormones causing increased metabolism. Causes include Graves' disease, toxic nodular goiter, and thyroiditis. Features: weight loss, heat intolerance, tremors, palpitations, anxiety, exophthalmos, and goiter.

Diagnosis: TSH, free T3/T4, thyroid antibodies, ultrasound, and radioactive iodine uptake scan.

Management: hypothyroidism treated with levothyroxine; hyperthyroidism treated with antithyroid drugs (methimazole, carbimazole, propylthiouracil), radioactive iodine, beta-blockers, and surgery (thyroidectomy) in selected cases.

Nursing care: monitor vital signs and weight; administer thyroid medications; teach medication timing (levothyroxine on empty stomach); monitor for thyroid storm (hyperthyroidism emergency) or myxedema coma (hypothyroidism emergency); provide rest and nutrition education.

8.3 Nursing Care Plan: Risk for Unstable Blood Glucose

Assessment

Fasting glucose 180 mg/dL, HbA1c 8.5%, polyuria, polydipsia, non-adherence to diet and medication.

Diagnosis

Risk for unstable blood glucose level related to inadequate diabetes management.

Interventions	Monitor blood glucose regularly; administer insulin/oral hypoglycemics; teach carbohydrate counting and meal timing; encourage exercise; inspect feet daily; educate about hypoglycemia recognition and treatment.
Rationale	Stable glycemic control prevents acute complications and long-term organ damage.